Unix　实验报告

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一

1、读取系统所用用户名并逐行输出,格式如"第*个用户:***."。

2、定义大小为 10 的数组，使用冒泡法对其由小到大排序。

3、Ping 命令扫描网络内存活主机，判断当前网络里，在线的 IP 有哪些?

```
tpx@tpx-virtual-machine:~$ vim quest1.sh
tpx@tpx-virtual-machine:~$ chmod 777 quest1.sh
tpx@tpx-virtual-machine:~$ ./quest1.sh
list of all usernames:
 USER NO. 1name:root
 USER NO. 2name:daemon
 USER NO. 3name:bin
 USER NO. 4name:sys
 USER NO. 5name:sync
 USER NO. 6name:games
 USER NO. 7name:man
 USER NO. 8name:lp
 USER NO. 9name:mail
 USER NO. 10name:news
 USER NO. 11name:uucp
 USER NO. 12name:proxy
 USER NO. 13name:www-data
 USER NO. 14name:backup
 USER NO. 15name:list
 USER NO. 16name:irc
 USER NO. 17name:gnats
 USER NO. 18name:nobody
 USER NO. 19name:systemd-network
 USER NO. 20name:systemd-resolve
```

```
USER NO. 54name:hadoop
2. pop sort :
ori array : 3 2 31 29 50 87 1 0 82 9
after array : 0 1 2 3 9 29 31 50 82 87
3.net scan
net :192.168.193
start 192.168.193.0/24...
online : 192.168.193.2
online : 192.168.193.128
goodbye
tpx@tpx-virtual-machine:~$ S
```

Shell 文件：

```
tpx@tpx-virtual-machine:~$ cat quest1.sh
#!/usr/bin/bash

#  all users name
echo  "list of all usernames: "
cat /etc/passwd | awk -F: '{print " USER NO. " NR "name:"$1 }'

#2 pop sort
echo " 2. pop sort :"
array=(3 2 31 29 50 87 1 0 82 9)
echo "ori array : ${array[@]}"
n=${#array[@]}
for ((i=0;i<n-1;i++)) do
        for ((j=0;j<n-i-1;j++)) do
                if [[ ${array[j]} -gt ${array[j+1]} ]]; then
                        tem=${array[j]}
                        array[j]=${array[j+1]}
                        array[j+1]=$tem
                fi
        done
done
echo "after array : ${array[@]}"
```

```
#3 net search
echo "3.net scan "
read -p "net :" prefix
echo "start ${prefix}.0/24..."
for i in {1..254}; do
    if ping -c 1 -W 1 "${prefix}.${i}" >/dev/null 2>&1; then
            echo "online : $prefix.$i"
    fi
done
echo "goodbye"
```

二

1、编写一个名为 factorial 的程序，其中定义阶乘函数 jc，在程序中递归调用它计算阶乘。

2、编写一个名为 iffile 程序，它执行时判断/bin 目录下 date 文件是否存在

3、编写一个名为 greet 的问候程序，它执行时能根据系统当前的时间向用户输出问候信息。设从半夜到中午为早晨，中午到下午六点为下午，下午六点到半夜为晚上。

1.factorial.sh 编写：

```
tpx@tpx-virtual-machine:~$ vim factorial.sh
tpx@tpx-virtual-machine:~$ ./factorial.sh
result of factorial 6:720
tpx@tpx-virtual-machine:~$ cat  factorial.sh
#!/usr/bin/bash

jc()
{
        if [ "$1" -eq 0 ];then
           echo 1
       else
               echo     $(( $1 * $(jc $(($1 - 1))) ))
        fi
}

res=$(jc 6)
echo "result of factorial 6:$res"
```

2.iffile.sh 编写：

```
tpx@tpx-virtual-machine:~$ vim iffile.sh
tpx@tpx-virtual-machine:~$ chmod 777 iffile.sh
tpx@tpx-virtual-machine:~$ ./iffile.sh
file /bin/data does exist
tpx@tpx-virtual-machine:~$ cat iffile.sh
#!/bin/bash

if [ -e "/bin/date" ];then
        echo "file /bin/data does exist"
else
        echo file "/bin/data does not exist"
fi
tpx@tpx-virtual-machine:~$
```

greet.sh 编写：

```
tpx@tpx-virtual-machine:~$ vim greet.sh
tpx@tpx-virtual-machine:~$ chmod 777 greet.sh
tpx@tpx-virtual-machine:~$ ./greet.sh
./greet.sh: line 4: [: missing `]'
good night
tpx@tpx-virtual-machine:~$ date
2025年 06月 15日 星期日 19:00:32 CST
tpx@tpx-virtual-machine:~$ cat greet.sh
#!/bin/bash

hour=$(date +%H)
if [ $hour -ge 0 -a $hour -lt 12];then
        echo "good morning"
elif [ $hour -ge 12 -a $hour -lt 18 ];then
        echo "good afternoon"
else
        echo "good night"
fi

tpx@tpx-virtual-machine:~$
```

三:

实验任务

1、写一个 shell 来判断用户所输入的参数是否是一个数字。

2、在 linux 里编写一个 shell 程序，在目录/userdata 下创建 50 个目录分别是 user1-user50，并设置个目录的权限。

3、编写一个名为 chname 的程序，将当前目录下所有的.txt 文件更名为.doc 文件。

4、编写一个名为 ninenine 的程序在屏幕上输出九九乘法表。

5、编写一个名为 chuser 的程序，执行中每隔 5 分钟检查指定的用户是否登录系统，用户名从命令行输入；

如果指定的用户已经登录，则显示相关信息。

6、编写一个名为 mini99 的程序，在屏幕上输出小九九乘法表。

7、编写一个名为 puteven 的程序，在屏幕上输出 0 到指定数之间的所有偶数。

1. isnumber.sh 编写与运行:

```
tpx@tpx-virtual-machine:~$ vim isnumber.sh
tpx@tpx-virtual-machine:~$ chmod  777 isnumber.sh
tpx@tpx-virtual-machine:~$ ./isnumber.sh 123
it is a number
tpx@tpx-virtual-machine:~$ ./isnumber.sh abs
it is not a number
tpx@tpx-virtual-machine:~$ cat isnumber.sh
#!/bin/bash

if [[ "$1" =~ ^[0-9]+$ ]];then
        echo "it is a number"
else
        echo "it is not a number"
fi
```

2.分别创建 50 个目录的 shell 文件:

```
tpx@tpx-virtual-machine:~$ vim  makedictionaries.sh
tpx@tpx-virtual-machine:~$ chmod 777  makedictionaries.sh
tpx@tpx-virtual-machine:~$ ./makedictionaries.sh
[sudo] password for tpx:
already created 50 dirs with mod 777
```

```
tpx@tpx-virtual-machine:~$ ls
datecp.txt  Downloads     GCA_000007325.1.ffn  iffile.sh           mpi     Pictures   quest4.sh  test3
Desktop     experiment4   greet.sh             isnumber.sh         Music   Public     snap       test5
Documents   factorial.sh  hello                makedictionaries.sh p1.ffn  quest1.sh  Templates  Videos
tpx@tpx-virtual-machine:~$ cd ../
tpx@tpx-virtual-machine:/home$ ls
abigale  another  daniel  dennis  hadoop  tpx
tpx@tpx-virtual-machine:/home$ cd ../
tpx@tpx-virtual-machine:/$ l
bin@   cdrom/  etc/       home/  lib32@  libx32@      media/  opt/   root/  sbin@  srv/      sys/   tmp/       usr/
boot/  dev/    etc.group  lib@   lib64@  lost+found/  mnt/    proc/  run/   snap/  swapfile  test/  userdata/  var/
tpx@tpx-virtual-machine:/$ cd userdata
tpx@tpx-virtual-machine:/userdata$ ls
user1   user13  user17  user20  user24  user28  user31  user35  user39  user42  user46  user5   user8
user10  user14  user18  user21  user25  user29  user32  user36  user4   user43  user47  user50  user9
user11  user15  user19  user22  user26  user3   user33  user37  user40  user44  user48  user6
user12  user16  user2   user23  user27  user30  user34  user38  user41  user45  user49  user7
tpx@tpx-virtual-machine:/userdata$
```

```
tpx@tpx-virtual-machine:~$ cat  makedictionaries.sh
#!/bin/bash

sudo mkdir -p /userdata
for i in {1..50}; do
        sudo mkdir -p /userdata/user$i
        sudo chmod 777 /userdata/user$i
done
echo "already created 50 dirs with mod 777"
```

3．chname.sh 编写

```
tpx@tpx-virtual-machine:~/chnameuse$ ls
tpx@tpx-virtual-machine:~/chnameuse$ touch t1.txt
tpx@tpx-virtual-machine:~/chnameuse$ touch tr.txt
tpx@tpx-virtual-machine:~/chnameuse$ touch uig.txt
tpx@tpx-virtual-machine:~/chnameuse$ ls
t1.txt  tr.txt  uig.txt
tpx@tpx-virtual-machine:~/chnameuse$ vim chname.sh
tpx@tpx-virtual-machine:~/chnameuse$ chmod 777 chname.sh
tpx@tpx-virtual-machine:~/chnameuse$ ./777 chname.sh
bash: ./777: No such file or directory
tpx@tpx-virtual-machine:~/chnameuse$ ./chname.sh
chname completed
chname.sh  t1.doc  tr.doc  uig.doc
tpx@tpx-virtual-machine:~/chnameuse$ cat chname.sh
#!/bin/bash

for file  in *.txt;do
        if [ -f "$file" ];then
                mv  "$file" "${file%.txt}.doc"
        fi
done
echo "chname completed"
ls
tpx@tpx-virtual-machine:~/chnameuse$
```

4.ninenine.sh 编写：

```
tpx@tpx-virtual-machine:~$ vim ninenine.sh
tpx@tpx-virtual-machine:~$ chmod 777  ninenine.sh
tpx@tpx-virtual-machine:~$ ./ninenine.sh
1x1=1
1x2=2   2x2=4
1x3=3   2x3=6   3x3=9
1x4=4   2x4=8   3x4=12  4x4=16
1x5=5   2x5=10  3x5=15  4x5=20  5x5=25
1x6=6   2x6=12  3x6=18  4x6=24  5x6=30  6x6=36
1x7=7   2x7=14  3x7=21  4x7=28  5x7=35  6x7=42  7x7=49
1x8=8   2x8=16  3x8=24  4x8=32  5x8=40  6x8=48  7x8=56  8x8=64
1x9=9   2x9=18  3x9=27  4x9=36  5x9=45  6x9=54  7x9=63  8x9=72  9x9=81
tpx@tpx-virtual-machine:~$ cat ninenine.sh
#!/bin/bash

for ((i=1;i<10;i++));do
        for((j=1;j<=i;j++));do
                printf "%dx%d=%d\t" $j $i $(($i*$j))
        done
        echo
done
```

5.chuser.sh 编写：

```
tpx@tpx-virtual-machine:~$ vim chuser.sh
tpx@tpx-virtual-machine:~$ chmod 777 chuser.sh
tpx@tpx-virtual-machine:~$ who
tpx      tty2         2025-06-15 18:40 (tty2)
tpx@tpx-virtual-machine:~$ ./chuser.sh tpx
user tpx has logined in
tpx      tty2         2025-06-15 18:40 (tty2)
```

```
tpx@tpx-virtual-machine:~$ cat chuser.sh
#!/bin/bash

if [ -z "$1" ];then
        echo "usage: $0 username"
        exit 1
fi

while true;do
        if who|grep -q "$1";then
                echo "user $1 has logined in"
                who | grep "$1"
        else
                echo "user $1 hasn't logined in"
        fi
        sleep 300
done
```

6.mini99 程序编写：

```
tpx@tpx-virtual-machine:~$ vim mini99.sh
tpx@tpx-virtual-machine:~$ chmod 777 mini99.sh
tpx@tpx-virtual-machine:~$ ./mini99.sh
1X1=1   1X2=2   1X3=3   1X4=4   1X5=5   1X6=6   1X7=7   1X8=8   1X9=9
2X1=2   2X2=4   2X3=6   2X4=8   2X5=10  2X6=12  2X7=14  2X8=16  2X9=18
3X1=3   3X2=6   3X3=9   3X4=12  3X5=15  3X6=18  3X7=21  3X8=24  3X9=27
4X1=4   4X2=8   4X3=12  4X4=16  4X5=20  4X6=24  4X7=28  4X8=32  4X9=36
5X1=5   5X2=10  5X3=15  5X4=20  5X5=25  5X6=30  5X7=35  5X8=40  5X9=45
6X1=6   6X2=12  6X3=18  6X4=24  6X5=30  6X6=36  6X7=42  6X8=48  6X9=54
7X1=7   7X2=14  7X3=21  7X4=28  7X5=35  7X6=42  7X7=49  7X8=56  7X9=63
8X1=8   8X2=16  8X3=24  8X4=32  8X5=40  8X6=48  8X7=56  8X8=64  8X9=72
9X1=9   9X2=18  9X3=27  9X4=36  9X5=45  9X6=54  9X7=63  9X8=72  9X9=81
tpx@tpx-virtual-machine:~$ cat mini99.sh
#!/bin/bash

for ((i =1;i<=9;i++)); do
        for((j=1;j<=9;j++)); do
                printf "%dX%d=%d\t" $i $j $(($i*$j))
        done
        echo
done


tpx@tpx-virtual-machine:~$
```

7.puteve.sh 程序编写：

```
tpx@tpx-virtual-machine:~$ vim puteve.sh
tpx@tpx-virtual-machine:~$ chmod 777 puteve.sh
tpx@tpx-virtual-machine:~$ ./puteve.sh 9
0
2
4
6
8
tpx@tpx-virtual-machine:~$ cat puteve.sh
#!/bin/bash

if [ -z "$1" ]; then
        echo "usage :$0 last number to detect"
        exit 1
fi

for ((i=0;i<=$1;i++)); do
        if (( i % 2 == 0 )); then
                echo $i
        fi
done

tpx@tpx-virtual-machine:~$
```

零:

1、非交互地执行脚本，两种执行方式：用 sh 命令执行脚本文件;给脚本文件添加执行权限，用./命令执行。

2、使用单引号、双引号、倒引号、反斜线。

3、操作课上讲述环境变量、内部变量

4、变量算术 expr、双小括号、$()操作。

5、实现用二分查找法在一组有序的数字中进行查找 shell 脚本。

```
tpx@tpx-virtual-machine:~$ vim question0.sh
tpx@tpx-virtual-machine:~$ bash question0.sh
env variable and inner variable
your username: tpx
working dictionary: /home/tpx
shfile name: question0.sh
PID: 96225
single : $USER will not  be analyzed
double : tpx will be analyzed
reversed: "tpx"name
expr result : 10+20=30
double brackets: 20 - 10 = 10
cal with dollars sign 10 * 20 = 200
search num 9 and show its address
we use array :2 3 4 9 11 13 16 17
our target  :9
succceed  address  is 3
```

```
tpx@tpx-virtual-machine:~$ cat question0.sh
#!/bin/bash
echo "env variable and inner variable "
echo "your username: $USER"
echo "working dictionary: $PWD "
echo "shfile name: $0"
echo "PID: $$"

echo  single : '$USER will not  be analyzed'
echo "double : $USER will be analyzed"
echo "reversed: \"$USER\"name"

a=10
b=20
sum=`expr $a + $b`
echo "expr result : $a+$b=$sum"
#diffvalue=0
((val =  b - a ))
echo "double brackets: $b - $a = $val"
res=$(( a * b ))
echo "cal with dollars sign $a * $b = $res"

array=(2 3 4 9 11 13 16 17)
target=9
```

```
echo "search num 9 and show its address"
echo "we use array :${array[@]}"
echo  "our target  :9 "
left=0
right=$(( ${#array[@]}-1 ))
found=0
while  [ $left -le  $right ]; do
        mid=$(((left + right)/2))
        current=${array[$mid]}

        if [ $current -eq $target ]; then
                found=1
                echo "succceed  address  is $mid "
                break
        elif [ $current -lt $target ]; then
                left=$((mid  + 1))
        else
                right=$((mid  - 1))
        fi
done
if [ $found -eq 0 ]; then
        echo "failed"
fi
```